

# Nathaniel Taylor

## Languages - Tools - Areas of Expertise

- typescript
- python
- web crawling/scraping
- docker/podman
- chrome extension development
- rhel linux administration
- lua
- sql queries
- postgresql
- go
- sql server administration
- C
- C++
- windows administration
- ARM embedded development

## Work History

**WICX Radio** System Administrator and Web Administrator (remote) November 2024 - current

- Implemented log-cleaning cronjobs to fix a periodically crashing Barix server, increasing uptime and decreasing maintenance hours
- Figured out a method for repairing faulty Mp3 headers, ensuring that programs play properly and decreasing maintenance hours for other members of the team

**MX Technologies** Software Engineer (remote) March 2023 - June 2024

- Wrote dozens of crawlers and reverse engineered dozens of banking APIs
- Created new containerized development environment with ability to install/run Chrome without root for India-based employees

**Valhalla** Backend Software Engineer (remote) August 2022 - September 2022

- Instructed the company on how to transition from a REST server to a GraphQL server, culminating in the successful production and release of the GraphQL server with less than 1 hour of client downtime
- Implemented a GraphQL server in Go/gqlgen designed to run in any serverless cloud computing environment (deployed to GCP Cloud Run in production) which reduced server maintenance expenses
- Worked closely with the client to understand the legacy data model, data operations, and desired end state; defined a GraphQL schema that fully replaced the REST schema with additional functionality that made available the full range of client data that was previously inaccessible

**Dark Owl** Software Engineer (hybrid) October 2021 - July 2022

- Created tools to more efficiently ingest messy data from the dark web, decreasing compute costs and increasing accuracy
- Wrote Python dark net crawler with a browser-based front end in Javascript for use by non-technical staff

**Picnic**   Software Engineer   (hybrid)   January 2020 - October 2021

- Created domain-specific markup language “Qform” for crawling search engine results
- Created “Userper”, a search engine crawler for multiple search engines
- Wrote major aspects of the terraform deployment infrastructure

**Center for Non-Proliferation Studies**   Open Source Intelligence Analyst   (on-site)   July 2015 - July 2018

- Created a 3D online "museum" of North Korea's ballistic missile program, in addition to numerous other 3D models for public engagement and media use
- Made significant contributions to the correctness and completeness of online resources, which explain highly technical topics for a lay audience

## Education

**Middlebury Institute of International Studies at Monterey**

**Master of Arts in Nonproliferation and Terrorism Studies**

July 2015 - May 2018

GPA: 3.7

**Virginia Commonwealth University**

**Bachelor of Arts in Political Science**

August 2010 - May 2014

GPA: 3.8